📖 User Manual – AMS Display

The AMS Display allows convenient management of filament spools directly on the device or via a web interface.

🖥️ Display Operation

🔹 Overview

- Displays all AMS slots (4 or 8)
- Each tile represents one slot
- Color indicates material / status

👆 Touch Functions

🔸 Short tap on a slot

Opens the **NumPad to subtract weight**

- Enter the desired amount in grams
- Press “OK” to confirm
- The value is subtracted from the spool

🔸 Long press on a slot

Opens the **spool detail view**

Shows:

- Spool ID
- Material / Type
- Color
- Brand
- Weight

🔸 **Inside the spool popup**

- ❌ **X** → closes the popup
- 🔁 **Swap** → opens the swap NumPad

🔁 Swap spools

1. Long press a spool
2. Press “Swap”
3. Enter target spool number
4. Confirm with “OK”

👉 The two spools will be swapped

🔢 NumPad Functions

- Numbers → input
- DEL → delete last digit
- OK → confirm
- X → cancel

🌐 WebUI Operation

The web interface can be accessed via the device’s IP address.

🔹 Overview

- Displays all spools
- Filters:
  - All
  - AMS (active slots)
  - Storage

🔹 Tile Interaction

Each tile has three main functions:

- **Tap on tile**
  → Opens input to **subtract weight**
- **“Swap” button**
  → Starts the **spool swap process**
- **“Edit” button**
  → Opens spool **data editing**

🔹 Edit Spool Data

Inside the edit popup you can modify:

- Material
- Type
- Color
- Article number
- Brand

🔹 “Empty” Function

- Enable the “Empty” checkbox
- The spool is marked as empty

👉 Display:

```
Empty (Spool X)
```

🔹 Import / Export

- 📤 Export → save spool list
- 📥 Import → load spool list

🔹 WiFi Setup

- Enter SSID and password
- Device restarts
- Connects automatically

📡 WiFi Behavior

- First start → Access Point (setup mode)
- After setup → connects to WiFi
- On failure → falls back to AP mode

📶 WiFi Status Display

- 🟢 Connected → shows WiFi name + IP
- 🔴 AP Mode → setup network active

🔗 MQTT Features (optional)

If MQTT is enabled:

- Sends:
  - Spool data
  - Weight values
  - Status information
- Can be integrated into:
  - Home Assistant
  - Node-RED
  - custom systems

🔹 Use Cases

- Monitoring filament usage
- Logging spool data
- Smart home integration

💡 Notes

- Slots and spools are independent
- Spools can be swapped freely
- Data is stored persistently (LittleFS)

⚠️ Tips

- Always confirm weight changes with OK
- Enter correct spool number when swapping
- WiFi credentials are case-sensitive

❤️ Support

For questions or issues:

- Use the GitHub repository
- Feedback is welcome